

Bag of Tricks for Training-Free Vision-Language-Action Models

What to See, When to Act, and How Much to Compute?

Guarded reuse



While the image and action tokens pass, the call is skipped, and the action is repeated, up to a cap.

Temporal fusion

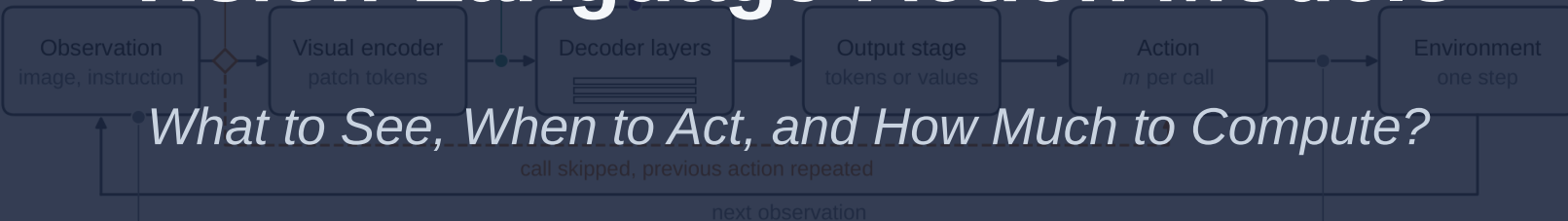


Unprotected patches keep their token. The fused token is up to the share of all patches.

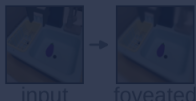
Depth pruning



The layers lowest in Block. If these are removed, none of the layers and the last kept.



Foveation



The image is blurred outside a central disc, the token count unchanged.

Action repeat



Each action is held for k steps with one call.

Anonymous submission · ICRA 2027 accompanying video

No trick is plug-and-play

Depth pruning on SimplerEnv WidowX, success change in points, three backbones.

CogACT

+9.5

50.0 → 59.5 %

SUCCESS

MiniVLA

-17.5

36.0 → 18.5 %

SUCCESS

SpatialVLA

-6.5

45.0 → 38.5 %

SUCCESS

The same trick helps one model, leaves another unchanged, and hurts a third.

Best setting per trick, Table I of the paper.

VLA models are strong but slow



CLIP: one original WidowX rollout

Overlay a live counter "ms per step" (e.g. CogACT 141 ms, SpatialVLA 424 ms).
Record from the evaluation run, 720p, 8-10 s.

Image + instruction



Vision encoder



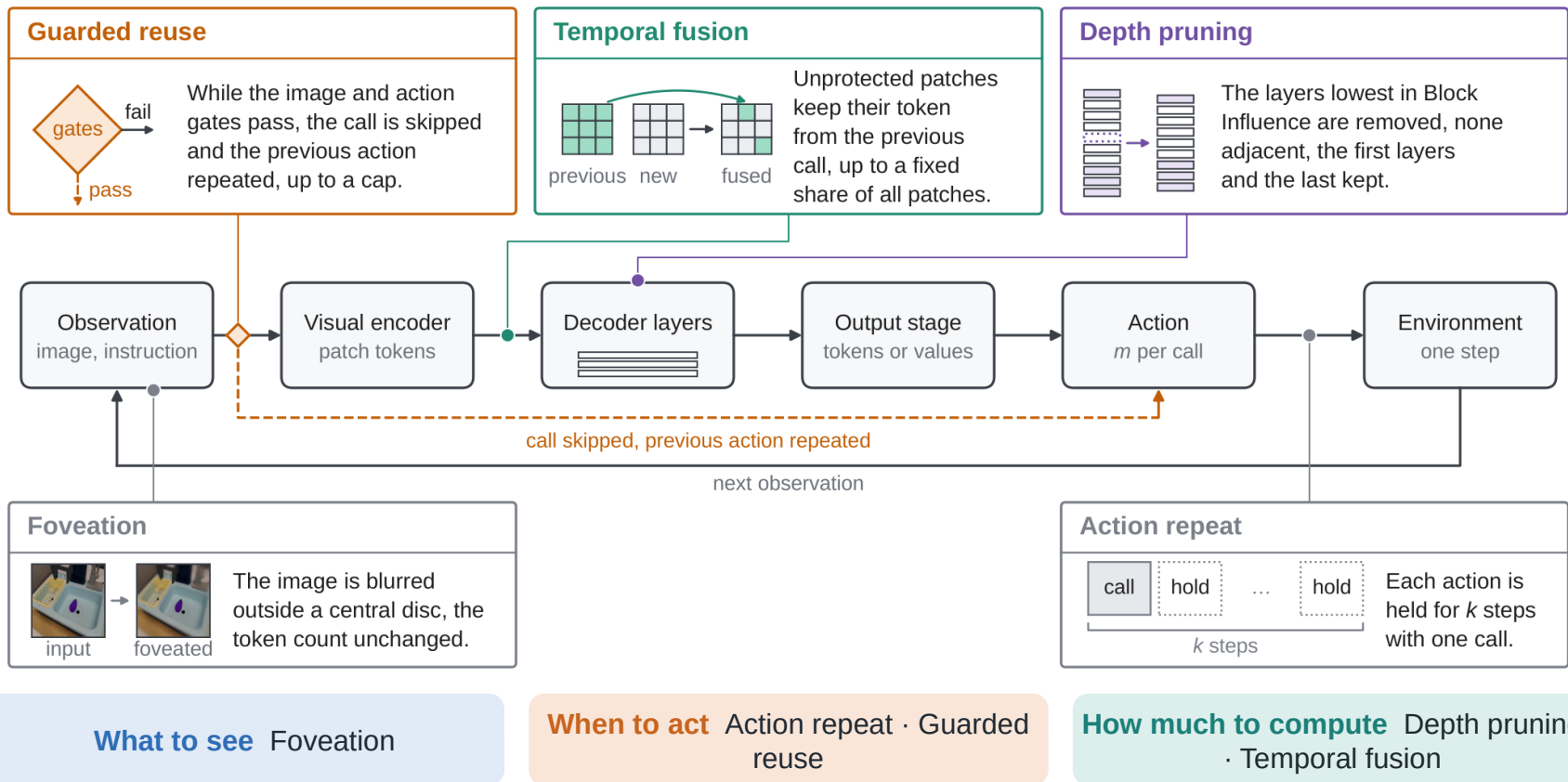
Language backbone



Action

- Every control step runs a large vision encoder and language backbone
- Existing tricks are each reported for one trick, one backbone, one benchmark
- Do the gains transfer?

Five tricks, three axes of the control loop



What to see: visual foveation

Foveation



input



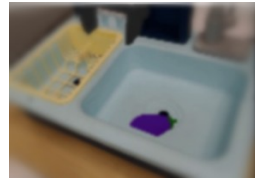
foveated

The image is blurred outside a central disc, the token count unchanged.

- Sharp central disc, progressively blurred periphery
- Keep ratio 20% or 50% of the image
- Token count unchanged, so this changes information, not FLOPs



Original frame



Keep 20%



Keep 50%

Tests whether a policy needs high-frequency detail only near the objects and the gripper.

When to act: action repeat vs guarded reuse

Original

policy call every step



Action repeat $k = 4$

one call, then three held actions



Guarded reuse

skip only while every gate passes

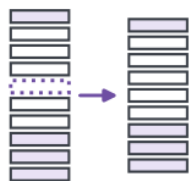


filled = policy call hollow = previous action held green dot = all gates passed

Gates: global and local image change small · recent actions agree · translation not near zero · gripper unchanged · reuse count below cap. Presets: strict, moderate, aggressive.

How much to compute: pruning and fusion

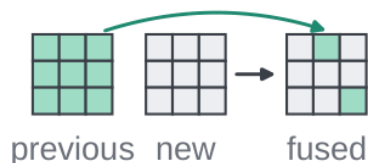
Depth pruning



The layers lowest in Block Influence are removed, none adjacent, the first layers and the last kept.

- Remove 1, 2, or 4 decoder layers with the lowest Block Influence
- None adjacent, first and last layers kept
- Selected once from calibration data, fixed before evaluation

Temporal fusion



Unprotected patches keep their token from the previous call, up to a fixed share of all patches.

- Reuse projected visual tokens of stable patches from the previous call
- Protect patches with motion, entropy, language attention, or interaction regions
- Presets: motion-entropy, task-aware, conservative-adaptive

Matched evaluation protocol

Backbone	WidowX	Fractal	LIBERO
CogACT (7.6B)	•	•	
OpenVLA (7B)	•	•	•
SpatialVLA (4B)	•	•	
CronusVLA (0.5B)	•	•	
UniVLA (8.5B)	•		•
MiniVLA (1B)	•		
SmolVLA (0.45B)			•

SimplerEnv WidowX: 4 tasks × 50 episodes. Google Robot/Fractal: 5 tasks × 50. LIBERO: Long, Goal, Object, Spatial.

22

backbone ×
environment pairs

13 + 1

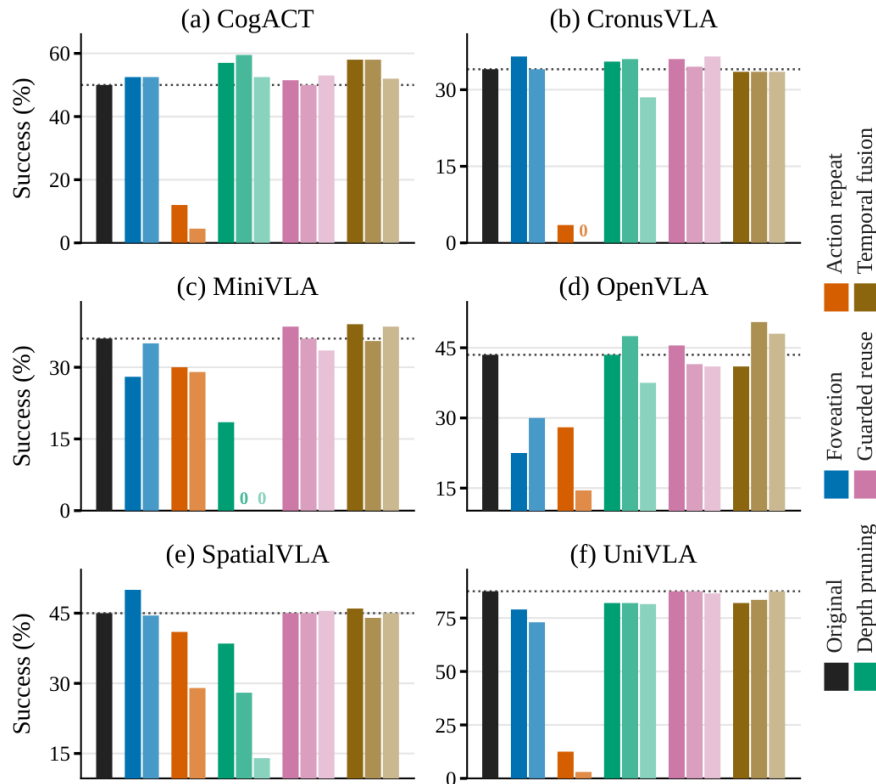
trick settings +
original, per pair

286

trick settings, all
paired with the
original

- Official checkpoints, no retraining
- Same task instances and initial states as the original
- Latency = wall-clock per environment step
- Exact two-sided McNemar test on paired episodes
- 4 × RTX 5090; tables report the best setting per trick

RQ-1 Which tricks preserve success?



A-1

Guarded reuse and temporal fusion match or improve success for most backbones.

Selected backbones also gain:

Foveation, WidowX: CogACT +2.5 · SpatialVLA +5.0 · CronusVLA +2.5

Depth pruning, WidowX: CogACT +9.5

Foveation, LIBERO Goal: UniVLA +7.0 vs SmoIVLA -13.0

Action repeat: mainly a speed baseline

Fig. 1: success per configuration, six WidowX backbones. Dotted line = original policy. Deltas from Tables I and II, best setting per trick.

Depth pruning helps CogACT on WidowX

Instruction

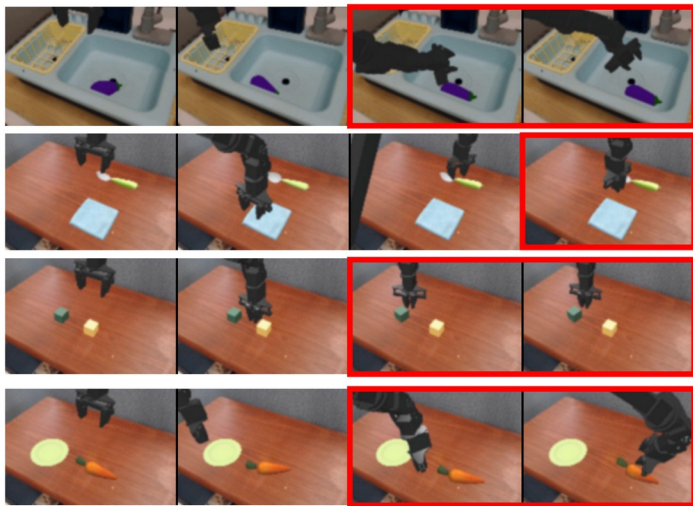
Put the eggplant in the basket.

Put the spoon on the table cloth.

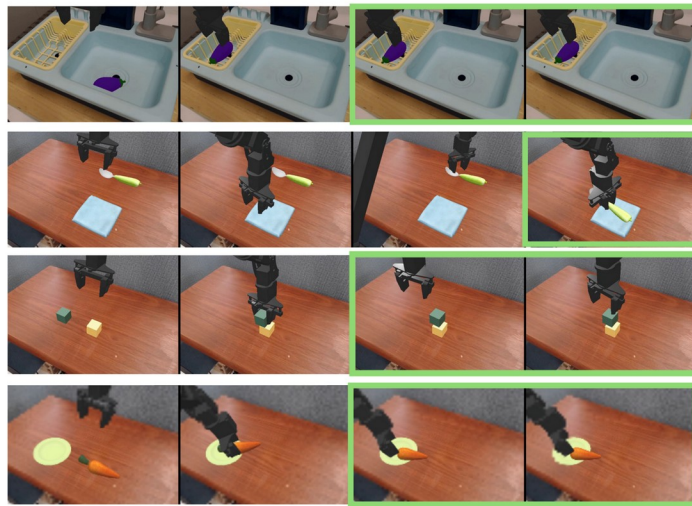
Stack the green cube on the yellow.

Put the carrot on the plate.

CogACT Original



CogACT + Depth Pruning



▶ CLIP: original vs depth pruning (paired episode)

Success 50.0 → 59.5 %

Latency 141.4 → 137.5 ms/step

Red = failure, green = success. Selected episodes, same task instances and initial states. Aggregate values from Table I.

Same trick, same benchmark, different backbone



CLIP: MiniVLA original

succeeds (green border)



CLIP: MiniVLA + pruning

fails (red border)

Selected episode, same initial state, WidowX

Depth pruning, WidowX

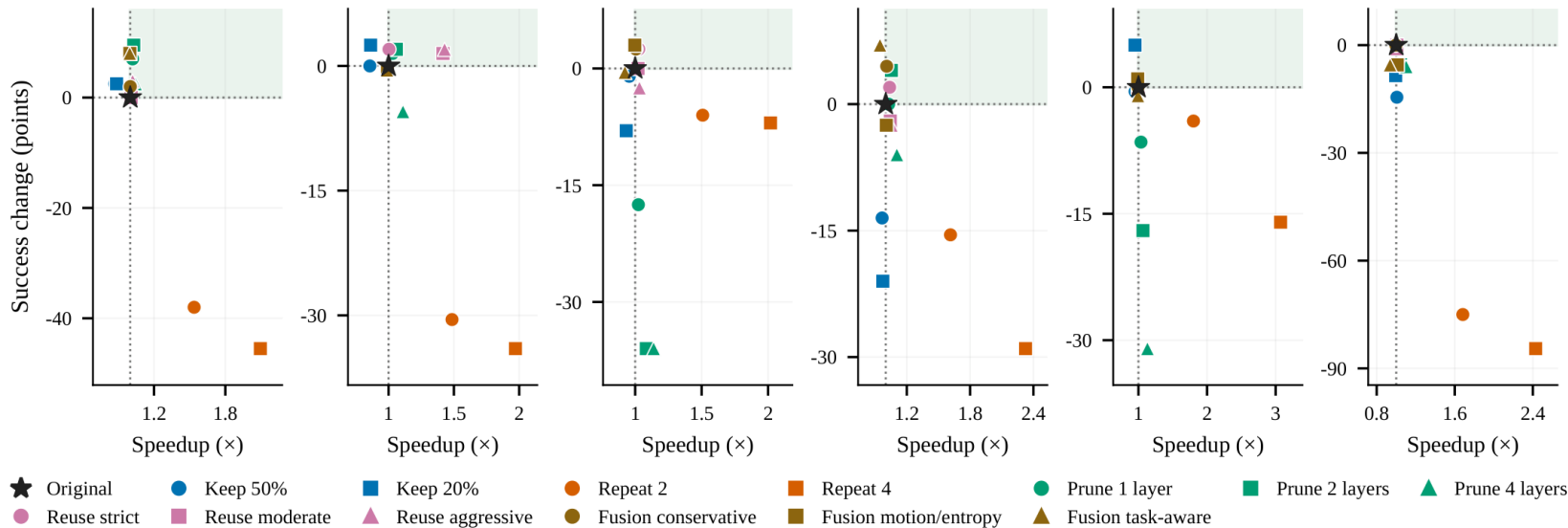
CogACT +9.5

MiniVLA -17.5

SpatialVLA -6.5

Structural redundancy is not a property of the trick alone. It depends on how each backbone uses its decoder layers during control.

RQ-2 Which tricks are actually faster?



A-2

Action repeat gives the largest speedup, depth pruning gives practical decoder-level savings, guarded reuse gives the safest trade-off.

Latency, ms per step	Original → trick
Repeat · SpatialVLA WidowX	423.5 → 235.1
Repeat · OpenVLA LIBERO Long	171.6 → 93.2
Pruning · OpenVLA WidowX	210.2 → 199.4
Reuse · SmoIVLA LIBERO Long	289.5 → 268.9

Open loop has a cost, gating keeps it safe



CLIP: Original

policy call every step

policy calls used: ____



CLIP: Action repeat $k = 4$

held actions, episode fails

policy calls used: ____



CLIP: Guarded reuse

gates shown, calls skipped only when they pass

policy calls used: ____

Action repeat, WidowX: CogACT -38.0 · UniVLA -75.0

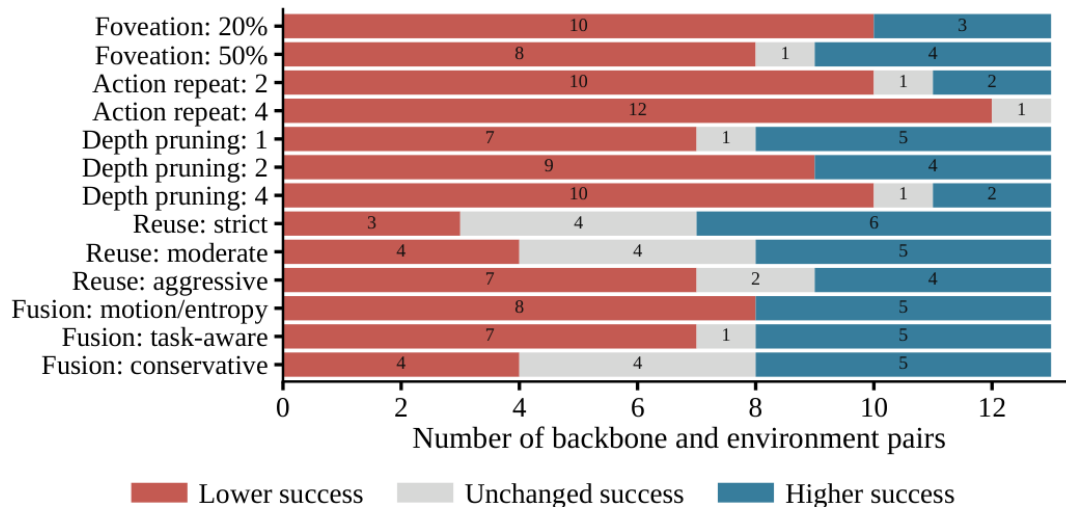
Guarded reuse, SmoIVLA LIBERO Long: 289.5 → 268.9 ms

Fractal, some backbones: CogACT -1.2 · SpatialVLA 0.0

success unchanged at 42.0 %

Same CogACT WidowX episode and initial state in all three panels. Deltas from Tables I and II, best setting per trick.

RQ-3 Does the effect transfer across models?



For each setting: how many of 13 backbone and environment pairs got lower, unchanged, or higher success (LIBERO suites pooled, so 13 stand for the 22 pairs).

A-3

Effects are a joint property of the trick, the backbone, and the task distribution.

Depth pruning, WidowX: CogACT +9.5 · MiniVLA -17.5

Foveation, LIBERO Goal: UniVLA +7.0 · SmoVLA -13.0

Action repeat: harmful on WidowX, milder on Fractal for some backbones

Not plug-and-play: each trick needs matched evaluation under the target backbone and environment.

Takeaways

Guarded reuse

Most reliable. Preserves success, modest speedup when the gates open.

Action repeat

Largest speedup, but the risk of long open-loop execution.

Depth pruning

Decoder-level savings, benefit depends on the backbone.

Foveation · Temporal fusion

Test how much the visual representation can change while the policy stays stable.

VLA efficiency is policy- and environment-dependent. Evaluate every trick under the target backbone and environment before applying it.

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Details, per-cell p-values, and all 286 settings are in the paper and its supplementary material.



Anonymous submission · ICRA 2027